Locating the Source of Hallucination in LLaVA-1.5:

A Two-Track Empirical Investigation of the ViT Visual Encoder

Progress Report — Tracks A and B Complete

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Abstract

Vision–language models (VLMs) hallucinate objects that do not appear in the input image. Most prior work treats this as a language-decoder failure and leaves the visual encoder unexamined. This project asks the opposite question: *is the ViT encoder already producing degraded representations on images where LLaVA-1.5 hallucinates?* We test this with three controlled experiments. Tracks A and B are now complete. **Track A** (attention-rollout entropy, n=500 COCO images) finds *no relationship* between [CLS] attention spread and hallucination outcome (ρ=0.003, p=0.94; Cohen's d=0.039). **Track B** (linear probes on patch embeddings at four ViT layers) finds the opposite: object discriminability is significantly lower on hallucinating images at every probed layer ($\Delta\text{acc} \in [-0.039, -0.027]$, Wilcoxon $p < 0.005$ for all four layers). Together these results refine the original hypothesis: the encoder failure is real, but it lives in *feature content*, not in *attention geometry*. This directly motivates Track C (causal patch ablation), which can now test whether feature-degraded images, identified by Track B, are exactly the ones where a ViT-attention-trusting steering method (VEGAS) is expected to fail.

1 Background and Hypothesis

The standard framing in object-hallucination research — benchmarks such as POPE [1] and CHAIR [2] — evaluates a VLM's textual output and treats the visual tower as a fixed black box. Recent work continues this pattern: VEGAS [4] steers generation using ViT attention as a trustworthy spatial signal, implicitly assuming the encoder is not the locus of failure.

Our central hypothesis inverts this assumption:

> **H0 (encoder-side hallucination):** On images where LLaVA-1.5 produces a hallucinated object, the ViT encoder's representations — attention maps and/or patch embeddings — are already measurably degraded relative to clean (non-hallucinating) images, *before* any token is generated.

The hypothesis decomposes naturally into three falsifiable sub-claims, each addressed by one track:

- **Track A.** Hallucinating images have more diffuse / less semantically focused [CLS] attention.
- **Track B.** Hallucinating images have less object-discriminative patch embeddings.
- **Track C.** Causally ablating "bad" patches changes hallucination behavior — confirming the encoder's role is causal, not merely correlational.

2 Shared Experimental Setup

- **Model.** `llava-hf/llava-1.5-7b-hf` loaded in `float16` via `AutoModelForImageTextToText`.
- **Encoder under study.** The CLIP ViT-L/14 visual tower internal to LLaVA-1.5 (24 layers, 1024-dim hidden, 24×24=576 patch tokens at 336×336 input).
- **Dataset.** COCO 2017 validation set, 500 randomly sampled images.
- **Hallucination ground truth.** CHAIR_i scoring against COCO instance annotations; an image is labelled *hallucinating* iff $\text{CHAIR}_i > 0$ (i.e., at least one mentioned object is not in ground truth).
- **Class balance.** 329 hallucinating / 171 clean (hallucination rate = 65.8%).
- **Compute.** Google Colab Pro on a single H100 (80 GB) for inference; CPU for probe training.

3 Track A — ViT Attention Rollout Analysis

Question

Does the ViT's [CLS] attention concentrate on semantically relevant patches in clean images, and does it become diffuse or misfocused in hallucinating images?

Procedure

1. For each of the 500 images, run a forward pass through the LLaVA ViT and capture per-layer attention tensors of shape $(H, 577, 577)$ for all 24 layers.
2. Compute attention rollout [3]: head-average each layer, add identity (residual), row-normalize, then take the matrix product across all 24 layers.
3. Extract the [CLS]-row of the rollout matrix (576 patch weights), and compute its Shannon entropy in nats. High entropy ⇒ diffuse attention; low entropy ⇒ focused attention.
4. Also compute spatial spread (normalized variance of patch coordinates weighted by the rollout distribution) and per-layer [CLS] entropy without rollout, for diagnostic purposes.
5. Stratify all measures by `is_hallucinating` and split into entropy quartiles (Q1 = most focused, Q4 = most diffuse).

Results

Test	Hallucinating	Clean	Effect / p	Verdict
n	329	171	—	—
Mean rollout entropy (nats)	6.3211	6.3207	$d = 0.039$	negligible
Mean spatial spread	0.4277	0.4252	$\Delta = 0.0025$	negligible
Spearman(entropy, CHAIR_i)	$\rho = 0.0033$		$p = 0.94$	null
Spearman(entropy, $\mathbb{1}$[hall])	$\rho = 0.0129$		$p = 0.77$	null
Welch t-test (entropy)	$t = 0.402$		$p = 0.69$	null
Mann–Whitney U (entropy)	—		$p = 0.39$	null

Table 1: Track A core statistics. No test reaches significance.

Quartile analysis confirms the same picture: the hallucination rate is non-monotonic across entropy quartiles — 61.6 % (Q1, most focused) → 68.8 % (Q2) → 68.8 % (Q3) → 64.0 % (Q4, most diffuse), a Q1→Q4 swing of only +2.4 pp. **Per-layer entropy curves** (layers 1–24)

show the largest hallucinating–clean gap at layer 19 ($\Delta = 0.130$ nats), but the difference is small relative to within-group variance and runs in the opposite direction at some layers.

Finding

The [CLS] attention rollout signal is statistically indistinguishable between hallucinating and clean images on this 500-image sample. **Track A returns a null result**: ViT attention diffusion does *not* predict hallucination at the global rollout level.

What this rules in / out

- **Rules out:** the simple version of H0 in which hallucination is driven by "the ViT looks at the wrong place."
- **Does not rule out:** that the encoder is the source. The information that distinguishes the two subsets may live in *what the patches encode*, not in *which patches are attended to*.

4 Track B — Patch Representation Linear Probing

Question

If attention geometry is uninformative, does the *content* of the patch embeddings encode object identity less reliably on hallucinating images?

Procedure

1. For each of the same 500 images, hook the ViT and capture hidden states at four probe layers: $L \in \{6, 12, 18, 24\}$.
2. For each layer, store both the [CLS] vector (1024-d) and the mean-pooled patch embedding (1024-d).
3. For each of the 80 COCO categories with ≥ 20 positive examples in our sample (17 categories survive: *backpack, bed, bench, book, bottle, bowl, car, cell phone, chair, cup, dining table, handbag, laptop, person, sink, truck, umbrella*), train a binary ℓ_2-normalized logistic regression probe (L2 penalty, `lbfgs`) under 5-fold cross-validation, predicting object presence from the mean-pooled patch embedding.
4. For each (layer, category) probe, compute three accuracies and three AUCs: overall, on the hallucinating subset only, and on the clean subset only.
5. Per layer, test $\Delta\text{acc} = \text{acc}_{\text{hall}} - \text{acc}_{\text{clean}}$ across the 17 categories using the Wilcoxon signed-rank test.

   A total of 4 layers $\times$ 17 categories = 68 probes were trained.

Results

Three observations.

1. Probe accuracy and AUC *rise with depth* and peak at layer 18 (acc = 0.946, AUC = 0.905), confirming that mid-late ViT layers carry the most object-discriminative information — consistent with prior probing work.
2. Probes are *uniformly worse on hallucinating images* than on clean images, at every layer. Layer 24 shows the largest absolute gap and the strongest significance ($p = 1.5 \times 10^{-4}$).

Layer	Mean Acc	Mean AUC	Acc_{hall}	$\text{Acc}_{\text{clean}}$	ΔAcc	(Wilcoxon p)
6	0.9086	0.7340	0.8954	0.9340	-0.0386	($p = 8.4\times10^{-4}$)
12	0.9326	0.8331	0.9204	0.9560	-0.0355	($p = 6.6\times10^{-4}$)
18	**0.9461**	**0.9049**	0.9369	0.9639	-0.0270	($p = 4.2\times10^{-3}$)
24	0.9388	0.8752	0.9260	0.9635	-0.0376	($p = 1.5\times10^{-4}$)
Mean ΔAcc across all four layers: -0.0347. All four layers significant at $\alpha = 0.005$.						

Table 2: Track B: linear-probe accuracy across ViT depth, stratified by hallucination outcome.

3. The ΔAcc is smallest at layer 18 (-0.027). The mid-late "peak" partially compensates, but does not erase, the encoder-quality gap.

Finding

Track B confirms an encoder-side signal of hallucination. Mean-pooled patch embeddings of hallucinating images are linearly less informative about object presence than those of clean images, with the effect significant ($p < 0.005$) at all four probed layers and largest at the input-near and output-near ends of the ViT.

5 Combined Conclusion from Tracks A + B

The two tracks together produce a sharper hypothesis than the one we started with.

> **Refined H0:** The ViT encoder *is* a measurable contributor to LLaVA-1.5 hallucination, but its failure does not appear as misdirected attention. It appears as degraded feature content — patch embeddings that carry less linearly-decodable object information — while the spatial attention distribution looks essentially identical to clean cases.

This has three consequences for the rest of the project:

1. **Methodological.** Any hallucination-mitigation method that trusts ViT attention as its corrective signal — VEGAS being the prime example — has a blind spot in exactly the regime where Track B says the encoder is failing. This is now an empirically grounded falsification target rather than a conjecture.
2. **Substantive.** Track B shows correlation, not causation. The same images that yield low probe accuracy might simply be intrinsically harder (more objects, smaller objects, rare categories). **Track C is required to convert the correlation into a causal claim.**
3. **Scientific contribution.** Even if Track C returns a partial or null result, the joint Track A null + Track B positive is itself a publishable finding: it relocates the encoder-failure signal from *attention* to *features* and explicitly identifies a class of methods (attention-trusting interventions) that cannot see it.

6 How Tracks A + B Set Up Track C

The original Track C plan called for ablating "patches corresponding to hallucinated object regions" identified by Track A attention maps. **Tracks A + B together force a more interesting design.** Because Track A's attention signal is uninformative, the natural patches to ablate are no longer "high-attention" or "low-attention" patches but rather *patches whose embedding content is degraded according to Track B*. This is a strictly stronger experimental design.

What Track C now tests

Track C central question: If we surgically modify the patch embeddings that Track B identifies as low-quality, does LLaVA's hallucination rate change in the predicted direction?

Proposed Track C procedure

1. **Patch selection (uses Track A + B outputs).** For each image, identify two patch sets: (i) *relevant patches*: those whose spatial location overlaps the COCO bounding box of the ground-truth object; (ii) *irrelevant patches*: those that do not. Track B's per-layer probe gives an additional axis: rank patches by their contribution to the probe logit (gradient × activation), giving us a "feature quality" ordering.
2. **Intervention 1: zero-ablation.** Replace selected patch embeddings with zeros at layer 18 (Track B's best layer) and re-run generation. Measure CHAIR_i change.
3. **Intervention 2: clean-image swap (counterfactual patching).** For an image A where LLaVA hallucinates object X, find a clean image B that does not contain X. Swap the patches of B corresponding to X's expected location into A. Does the hallucination disappear?
4. **Intervention 3: VEGAS falsification.** Apply VEGAS-style attention steering to the subset of images flagged as low-probe-accuracy by Track B. If VEGAS reduces hallucination on the high-probe-accuracy subset but *not* on the low-probe-accuracy subset, we have the falsification result the project was designed to produce.

Validation strategy for Track C

The intervention is meaningful only if controlled. The validation plan, per the original RESEARCH_PLAN.md and the new evidence:

- **Random-patch control.** For every targeted ablation, run a matched ablation that drops the same number of patches chosen uniformly at random. The targeted ablation must produce a significantly larger CHAIR_i change than the random-patch baseline.
- **Sham layer.** Run the same ablation at a layer Track B identifies as low-discriminability (layer 6). The effect should be weaker there than at layer 18, otherwise the result is not localized to the layers that carry object identity.
- **Reversal test.** Swap the labels: ablate "high-quality" patches in clean images. If clean images now hallucinate at a higher rate, the causal direction is confirmed.
- **Pre-registered effect size.** A paired Wilcoxon test on per-image CHAIR_i before/after intervention, with a minimum detectable effect of $\Delta\text{CHAIR}_i = 0.05$, $\alpha = 0.01$, two-sided.

What completing Track C would prove

1. **If targeted ablation reduces hallucination significantly more than random ablation:** the Track B correlation is causal, and the ViT encoder is established as a causal contributor to LLaVA-1.5 hallucination — which directly supports the project's core hypothesis and its publication framing ("isolating the encoder as a measurable cause").
2. **If clean-image patch swaps transfer hallucination outcomes:** hallucination travels with the patch features, not with the global image. This is the strongest possible evidence that the failure is encoded in patch content.

3. **If VEGAS works on high-probe-accuracy images but fails on low-probe-accuracy images:** the project delivers a clean falsification of the VEGAS assumption that ViT attention is universally trustworthy — exactly the contribution the proposal targets.
4. **Even a partial Track C result** (e.g., zero-ablation works, swap does not) is publishable as a workshop paper: the controlled experimental design and the Track A-null/Track B-positive dissociation are themselves a contribution.

7 Status

- Track A: **complete**. Outputs: `entropy_ranks.csv`, `rollout_data_complete.npz`, four figures, 22 attention-overlay visualizations.
- Track B: **complete**. Outputs: `patch_embeddings_cache.npz` (16 layer-feature matrices), `probe_results.csv` (68 probes), `probe_stats.csv`, four figures.
- Track C: design ready, gated on team-call task split. ETA two weeks.

References

[1] Li, Y. et al. (2023). *Evaluating object hallucination in large vision-language models.* EMNLP.

[2] Rohrbach, A. et al. (2018). *Object hallucination in image captioning.* EMNLP.

[3] Abnar, S. & Zuidema, W. (2020). *Quantifying attention flow in transformers.* ACL.

[4] VEGAS: ViT-attention-guided steering for VLM hallucination mitigation, arXiv:2512.12089.

[5] Dosovitskiy, A. et al. (2021). *An image is worth 16×16 words.* ICLR.